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Task complexity and AI dependency among college students: the mediating roles of cognitive load, future anxiety, and task motivation

Hanhui Li[1], Fan Yang[1*] and Jinpeng Liu[2]

Abstract

Purpose This study aims to clarify the associations between task complexity and Artificial Intelligence (AI) dependency among university students and to examine the roles of cognitive load, future anxiety, and task motivation in these associations.

Methods A structured questionnaire survey was conducted with 442 college students, collecting self-reported data on task complexity, AI dependency, cognitive load, future anxiety, and task motivation. Confirmatory factor analysis was used to assess the adequacy of the proposed framework, and structural equation modeling was then employed to examine the associations among the variables.

Results Task complexity was significantly and positively associated with AI dependency. Further analyses indicated that students facing more complex tasks tended to report higher cognitive load and stronger future anxiety, which were linked to greater AI dependency. In contrast, students with higher task motivation were less likely to over-rely on AI even when dealing with complex tasks. No significant gender differences were found in AI dependency or related psychological variables.

Conclusion This study advances a task-focused perspective on AI dependency by integrating Cognitive Load Theory and Self-Determination Theory and by specifying cognitive load, future anxiety, and task motivation as correlates linking task complexity with AI dependency. Practically, the results suggest actionable directions for course and assessment design (e.g., calibrating task complexity, scaffolding task decomposition and staged feedback, and supporting motivation and emotion management) to help students balance AI use with autonomous learning.

Keywords Task complexity, AI dependency, Cognitive load, Future anxiety, Task motivation

*Correspondence:
Fan Yang
yangfan_fzfu_edu@163.com
[1]School of Foreign Languages, Fuzhou University of International Studies and Trade, Fuzhou, Fujian 350202, China
[2]International College, Fujian Agriculture and Forestry University, Fuzhou, Fujian 350007, China

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Introduction

Given how quickly AI technology is developing, its use is ubiquitous across various fields [1, 2]. The use of AI in higher education has significantly increased college students' learning effectiveness and provided personalized learning experiences through intelligent support systems [3]. However, as AI technology becomes more widespread, the issue of AI dependency (AID) has gradually emerged, becoming a crucial factor related to students' learning and mental health [4–6]. AID refers to the over-reliance on AI technology in many areas of life (including social interactions, daily work, and academic studies), which is marked by a lack of creative thinking and reduced decision-making autonomy [7]. Studies have shown that individuals with AID often display laziness and impatience when faced with tasks, preferring to rely on AI to quickly complete tasks [8, 9]. Therefore, understanding the factors associated with AID is important not only for clarifying its potential educational and psychological implications but also for describing how college students relate to AI in their everyday learning.

A number of variables are associated with the development of AID, with individual characteristics and TC playing key roles. First, individual traits such as academic self-efficacy, personality traits (e.g., self-critical perfectionism and negative academic emotions), future anxiety (FA), and cognitive abilities are significantly related to students' reliance on AI [10–14]. For example, study conducted by Wijaya et al. [11] indicates that individuals who believe they are not as capable of succeeding academically tend to report greater reliance on AI when managing academic stress and negative emotions. Gerlich [14] also found that students with lower cognitive abilities are more inclined to delegate cognitive tasks to AI, which is associated with reduced involvement in deep thinking and independent analysis, thereby relating to higher levels of AID. In addition, prior studies on AI-supported learning have shown that students' attitudes toward AI and their motivation for using it are closely related to their patterns of technology use [15, 16]. Recent studies suggest that students who perceive AI tools as useful, trustworthy, and easy to control generally report more frequent use and are more likely to turn to them when facing academic workload or time pressure, whereas stronger skepticism and ethical or privacy concerns are associated with more cautious or limited use of such tools [17–19]. Similarly, students who are motivated to use AI primarily to reduce effort or to cope with time pressure may be more prone to developing dependency, compared with those who use AI mainly as a supplementary resource to support autonomous learning [20, 21]. In addition, the convenience and efficiency of AI, particularly its support for personalized learning, appear to encourage students' unconscious development of dependency on technology, especially in highly digitized educational environments [22]. Apart from individual factors and technological environments, task characteristics, particularly task complexity (TC), are also closely linked to AID [23]. These findings suggest that AID reflects not only the presence of AI tools but also how students evaluate them and why they choose to use them in their daily academic work, alongside the characteristics of the tasks they need to complete.

While research on AI-supported learning is expanding, task-related characteristics have received comparatively less attention in studies on AID. Existing work tends to treat the use of AI as an individual choice driven by personal traits or perceived usefulness, whereas the specific demands of academic tasks and their psychological consequences are rarely examined in a unified framework. From a task perspective, complex assignments may be associated with higher cognitive load (CL), more intense future-oriented worry, and shifts in task motivation (TM), all of which could be related to students' tendency to rely on AI tools. Consistent with previous research [24, 25], CL is conceptualized as the mental effort and cognitive resources that students need to invest to process task information and complete the required steps, FA reflects worry and uncertainty about future academic and career outcomes, and TM captures students' willingness and reasons to invest effort in a particular learning task. Prior studies have also highlighted these constructs as targets for educational and psychological interventions [26–29], such as reducing unnecessary task demands and redundant information to help students manage CL, providing academic and career guidance to address FA, and using autonomy-supportive and competence-enhancing teaching practices to strengthen TM. Yet empirical studies that simultaneously consider these cognitive, emotional, and motivational processes in relation to AID among college students remain scarce. To address this gap and extend prior work on these constructs, the present study focuses on TC and examines how it is related to AID through CL, FA, and TM, while also exploratorily investigating potential gender differences in these associations.

Theoretical framework

This study investigates the relationships between TC, CL, FA, TM, and college students' AID in the context of academic tasks, from an integrated perspective of Cognitive Load Theory and Self-Determination Theory (SDT). Cognitive Load Theory [24] emphasizes that individuals are limited by the capacity of working memory when completing tasks. The higher the demands of a task in terms of information processing, the number of operational steps, and the integration of rules, the greater the occupation of limited cognitive resources. Previous research has shown that Cognitive Load Theory is highly

applicable and provides strong theoretical explanations for learning burden, academic stress, and resource allocation in technology-supported learning environments [30–32]. Within this framework, TC is regarded as the situational level of task demands, reflecting the challenges that tasks pose to learners in terms of information volume, clarity of structure, and uncertainty in goals and evaluation criteria. CL, on the other hand, describes individuals' subjective cognitive experience when performing tasks, including resource consumption, mental fatigue, and operational difficulties. In this study, TC and CL are positioned as core indicators of task demands and cognitive resource utilization in academic tasks, providing a cognitive basis for understanding students' AID in high-demand task situations.

SDT focuses on the fulfillment of basic psychological needs during the behavior process, positing that the degree of autonomy, competence, and relatedness satisfaction is closely related to the quality of motivation [26]. Existing research shows that SDT has been widely applied in fields such as learning motivation, academic engagement, and educational technology use, with TM and emotional responses related to academic outcomes often regarded as important psychological components for understanding students' behavioral choices in educational contexts [27–29]. In academic settings, TM reflects students' subjective willingness and behavioral reasons for engaging in learning tasks, which can manifest as relatively autonomous motivation based on interest and personal values, or controlled motivation driven more by external demands and performance pressure. FA is an emotional experience directed at future outcomes, reflecting individuals' worries and uncertainty about academic performance, graduation prospects, or career development. In this study, TM and FA are positioned as motivational and emotional background variables within the task context, used to describe students' subjective engagement and future-oriented experiences when facing academic demands. Based on the two theories mentioned above, this study constructs a conceptual framework that integrates situational demands, internal psychological processes, and behavioral outcomes. It examines TC, CL, TM, FA, and AID under the same task perspective. This integrated framework provides a theoretical basis for the subsequent hypotheses on the relationships among TC, CL, FA, TM, and AID, as well as offering a relatively systematic psychological perspective for describing the relationship between college students and AI tools in academic tasks. As demographic variables are not part of the proposed theoretical mechanisms, gender is included exploratorily as a between-group factor to test the robustness of the focal task-related pathways rather than as a theoretically derived moderator.

Review and hypotheses

Task complexity (TC) and AI dependency (AID) TC refers to the CL, difficulty in information processing, and the intricacy of task steps that an individual faces while completing a task [33]. High information volume, high error rates, and uncertainty are frequently characteristics of complex activities, which require individuals to invest more cognitive resources during task execution [34]. Higher TC is associated with greater cognitive resource demands, which are linked to increased CL and potentially to task efficiency and completion quality [35, 36]. For example, Asgari et al. [36] noted that clinical practitioners managing complex tasks are often required to integrate large amounts of information and make multifaceted decisions, which are associated with substantial CL. In such contexts, individuals facing complex tasks tend to rely on external tools, aiming to reduce CL and enhance task efficiency. According to Cognitive Load Theory [24], when task demands exceed an individual's cognitive capacity, cognitive bottlenecks may occur, which may be associated with decreased task performance efficiency and may be accompanied by psychological stress and negative emotions. In such situations, individuals may seek external assistance to manage CL, and AI is often perceived as an effective option. AI tools are regarded as providing automated support that can help individuals handle complex tasks more efficiently. Research has shown that when tasks are perceived as unappealing or overly complex, individuals tend to exhibit a stronger reliance on AI, viewing it as a means to ease the task burden and enhance efficiency [37, 38]. For instance, Lu and Liu [37] discovered that when task design was complex and required multi-step decision-making, students reported significantly higher dependence on intelligent recommendations, content summarization, and planning tools in AI-assisted learning systems. Taken together, these studies suggest that when students perceive academic tasks as more complex, they are more likely to turn to AI tools to help them manage the demands of these tasks and to reduce their effort.

Mediating role of cognitive load (CL) CL refers to the mental effort and cognitive resources involved in processing a task and is associated with factors such as TC, information volume, and individual cognitive abilities [25]. According to Cognitive Load Theory [24], when a task's cognitive requirements surpass an individual's processing capacity, CL tends to increase, which may be associated with the efficiency of information processing and task completion quality. Studies have shown that higher CL is often associated with individuals being more likely to seek external tools, such as AI technology, to alleviate the psychological burden of tasks and enhance efficiency [39–42]. For instance, Doger and Gocen [42] pointed out

that while AI helps teachers handle complex tasks and eases their workload, it may also lead to overreliance on AI tools. This suggests that in situations with high cognitive demands, individuals are more likely to depend on AI to accomplish tasks.

Within the framework of Cognitive Load Theory, TC is closely associated with the magnitude of CL, as more complex tasks are linked to greater demands on cognitive resources for processing, integration, and decision-making [24]. As an example, Zeitlhofer et al. [43] discovered a positive correlation between TC and CL, indicating that tasks with higher complexity tend to correspond to higher levels of CL, which may, in turn, relate to task efficiency and completion quality. Similarly, Liu et al. [44] conducted a study with 128 undergraduates and found that multimedia presentations with high interactivity (complexity) were associated with more challenging tasks and had the potential to cause cognitive overload. Therefore, we expect that CL helps to explain why students facing more complex tasks report higher AID.

Mediating role of future anxiety (FA) FA is an emotional experience rooted in concerns about the uncertainty of the future, typically manifesting as worry about potential negative events and fear of unknown outcomes [45]. When individuals face future situations, they often feel anxious due to a lack of control and foresight, and this emotional response may be associated with their decision-making process and behavioral patterns [46]. In the context of the increasing integration of AI technology, FA is often associated with a greater tendency to seek external technological support, particularly when dealing with uncertain or challenging tasks [47–50]. For example, Fan et al. [49] found that individuals facing uncertainty about the future and challenging tasks tended to experience higher levels of FA, which was associated with a stronger likelihood of seeking external technological support, particularly AI, for immediate feedback and to reduce decision-making pressure, thereby alleviating concerns about potential future failure. Additionally, according to Ni and Cheung [48], higher FA levels among college students were associated with a higher likelihood of rely on AI tools like automated assignment generators and intelligent learning recommendation systems to provide them with temporary psychological security and control when making academic decisions.

There is a close relationship between the generation of FA and TC. While research on the direct relationship between the two is limited, current research indicates that TC is positively associated with an individual's anxiety [51, 52]. According to Cognitive Load Theory [24], when individuals face complex tasks, they must allocate more cognitive resources to process information and make decisions. As TC increases, CL also tends to rise, and when it surpasses an individual's processing capacity, anxiety levels may become higher. As an example, Jebur et al. [52] discovered a strong positive association between TC and anxiety, specifically noting that as TC increases, specifically noting that higher TC was linked to significantly greater anxiety, especially when the task challenges exceed their skill level. Based on this finding, it can be hypothesized that as college students face increasingly complex academic tasks and uncertainties regarding their future careers, their FA may also tend to rise. Therefore, FA, as an emotional response associated with students' experiences of TC, warrants further investigation.

Mediating role of task motivation (TM) TM is the internal or external motivation associated with a person's preparedness, diligence, and persistence in finishing a particular activity [53]. Studies have shown that TM is significantly associated with individual's performance and behavior when facing challenges [54]. In particular, when individuals report insufficient TM in learning or work-related tasks, they tend to invest less effort and are more likely to rely on assistive tools to complete the task [55]. Although current research has not directly explored the relationship between TM and AID, existing studies provide indirect evidence supporting this connection. For example, Chuang et al. [55] found that college students with high TM exhibited greater autonomy and persistence in their learning. Similarly, Frenkenberg and Hochman [56] suggested in a study on AI usage that when individuals perceived higher autonomy in a task, their dependency on AI was significantly lower. Therefore, it can be inferred that the degree of TM is intimately associated with an individual's reliance on AI technology.

TC and a person's TM are closely connected, particularly in learning or work-related tasks. According to SDT [26], situations with higher TC are often associated with individuals perceiving the task as too difficult, which may influence their TM. Let's say, Masrom et al. [57] found in a study on the relationship between lexical complexity in language production and TM that when tasks require higher causal reasoning or have complex structures, learners' TM tended to show a negative correlation. Additionally, Soh [58] observed in a study on college students' writing conditions that increased TC was associated with lower TM, particularly when the tasks became too difficult, which was linked to decreased interest and participation. Therefore, we hypothesize that lower TM partly explains why higher TC is associated with stronger AID.

Exploratory consideration of gender differences

Direct research on gender differences in AID is limited and has produced mixed findings. Some studies suggest

that males and females may differ in their patterns of technology adoption, interaction styles, and risk perception, whereas others report no consistent gender gap in technology-related outcomes [59–61]. It is therefore unclear whether such differences extend to AID and to the task-related psychological variables examined in this study. Given these inconsistencies, we treat gender as an exploratory between-group factor and test whether there are mean differences in AID and in its task-related psychological correlates (CL, FA, TM) between male and female students.

Aims and hypotheses

Based on the above review, this study adopts a task-focused perspective to examine how perceived TC is associated with college students' AID in academic learning. Guided by cognitive load theory and SDT, we test whether CL, FA, and TM help explain this association and exploratorily assess potential gender differences in these mechanisms.

Key research questions

RQ1: How is TC associated with AID in academic tasks among college students?
RQ2: To what extent do CL, FA, and TM account for the association between TC and AID?
RQ3: Are there gender differences in AID and in the task-related cognitive, emotional, and motivational processes linking TC with AID?

Hypotheses

H1: TC is positively associated with AID.
H2: CL mediates the association between TC and AID.
H3: FA mediates the association between TC and AID.
H4: TM mediates the association between TC and AID.
H5: Gender moderates the proposed relationships between TC, CL, FA, TM, and AID.

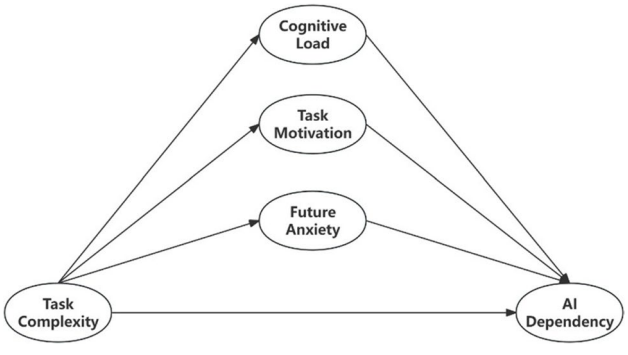


Fig. 1 Research model

The research model is shown in Fig. 1:

Method

Procedure

This research received ethical approval from the Ethics Committee of Fuzhou University of International Studies and Trade. All participants provided informed consent, and the study complied with institutional ethical guidelines, safeguarding students' privacy and autonomy. Data were collected between February and April 2025 using an anonymous online questionnaire administered via Questionnaire Star (www.wjx.cn). To recruit participants, we used random sampling among full-time undergraduate students at several universities in China. With the assistance of student affairs staff and course instructors, classes were randomly selected from the undergraduate population, and the survey link was distributed to these classes. Students who were willing to participate accessed the link on their personal devices, read an online information page describing the study aims, the voluntary nature of participation, anonymity and confidentiality, and their right to withdraw at any time, and then indicated their consent before completing the questionnaire. The questionnaire used in this study contained 39 items. According to the rule of thumb of at least 10 participants per item and assuming an approximate 20% potentially invalid responses, the minimum required sample size was estimated to be approximately 468 participants (39 items × 10 participants + 20% × 39 items × 10 participants) [62]. In total, 471 questionnaires were initially returned. Predefined criteria were then applied to exclude invalid surveys: (1) questionnaires with more than five missing answers (legitimate questionnaires had at least 80% of the items completed; questionnaires with more than 20% missing responses were deemed invalid); (2) questionnaires in which 80% or more of the items were rated as "strongly agree" or "strongly disagree", in order to avoid serious floor or ceiling effects and ensure the accuracy of the data analysis; and (3) implausibly short completion times, defined as less than one-third of the median completion time, which indicate insufficient effort responding [63, 64]. After applying these criteria, a valid dataset of undergraduate respondents was retained and used to test the hypothesized model.

Participants

Participants were full-time undergraduate students from several universities in China. To ensure a homogeneous academic stage and improve interpretability, the study intentionally focused on undergraduates only; within this population, participants were classified by grade (freshman, sophomore, junior, and senior) to capture potential differences in academic tasks and coursework complexity that may influence students' reliance on AI. After data

screening according to the predefined exclusion criteria, 442 valid questionnaires were retained, resulting in a final sample of 267 male students (60.5%) and 175 female students (39.5%). With respect to grade level, 95 students were freshmen (21.4%), 151 were sophomores (34.2%), 77 were juniors (17.4%), and 119 were seniors (27.0%). In terms of age, 136 participants were younger than 18 years (30.8%), 235 were between 18 and 22 years (53.1%), and 71 were older than 22 years (16.1%). The demographic characteristics of the participants are presented in Table 1. The sample distribution did not significantly differ from the overall population distribution, indicating that the sample was highly representative, and gender, grade level, and age were not significantly associated with the study variables.

Measurement tools

There were two sections in the questionnaire: the constructs used in this investigation and the demographic data of the participants. Before the formal survey, we conducted a small pilot test with 20 undergraduate students from the intended audience to guarantee the clarity of the modified scale elements, relevant, and easy to understand. Minor wording refinements were made based on their feedback without altering the original constructs [65]. The study variables were measured using well-established scales, with slight wording adjustments to fit the specific context of this research.

Task complexity (TC), defined as students' perceived difficulty and complexity of their academic tasks when using AI tools, was measured using the NASA-TLX (Task Load Index) scale [66]. The scale includes 5 items in 1 dimension (e.g., "My work tasks involve many different elements, and AI helps me integrate these elements and provide clear solutions"). A 5-point Likert scale was used, ranging from 1 to 5, with intermediate values reflecting varying levels of agreement. The scale used in this investigation had a consistent internal consistency, as indicated by its Cronbach's alpha coefficient of 0.890.

Cognitive load (CL), referring to the mental effort and cognitive strain experienced when completing AI-supported academic tasks, was measured using the Cognitive Load Scale [67]. The scale contains 4 items (e.g., "It is difficult for me to predict the answers in the AI-generated 3D learning resources, but it is also a challenge") and consists of 1 dimension. A 5-point Likert scale was used, ranging from 1 to 5, with intermediate values reflecting varying levels of agreement. With a Cronbach's alpha coefficient of 0.893, the scale employed in this study demonstrated consistent internal consistency.

Future anxiety (FA), conceptualized as worry and negative expectations about one's academic and career future, was assessed using the Dark Future Scale [45]. The scale includes 5 items (e.g., "I fear that the problems troubling me now will last for a long time") and consists of 1 dimension. A 5-point Likert scale was used, ranging from 1 to 5, with intermediate values reflecting varying levels of agreement. A higher score indicates higher FA. The Cronbach's alpha coefficient for the scale in this study was 0.920, indicating stable internal consistency.

Task motivation (TM), that is, students' willingness and reasons to engage in academic tasks when AI tools are available, was measured using the Task Motivation Scale [68]. The scale contains 16 items (e.g., "I enjoy tasks because AI provides interesting content that makes learning more attractive") and is divided into 6 dimensions: General Preferences, Future Influence, Content and Novelty, Efforts, Diagnosis, and Practice Value. A 5-point Likert scale was used, ranging from 1 to 5, with intermediate values reflecting varying levels of agreement. A higher score indicates stronger TM. The scale used in this study had a Cronbach's alpha coefficient of 0.962, indicating great internal consistency.

AI dependency (AID), defined as students' tendency to rely excessively on AI tools to complete academic work and make learning-related decisions, was measured using the five-item AI Dependence Scale reported by Huang et al. [23]. In their study, five items were adapted from established smartphone addiction/problematic use measures to reflect reliance on AI. Huang et al. reported good psychometric properties for this scale, including high internal consistency (Cronbach's α = 0.881 at T1 and 0.953 at T2) and adequate construct validity supported by confirmatory factor analysis (CFI = 0.955–0.992; TLI = 0.980–0.987). In our study, we used the same five items without any further modification. Participants rated each item on a 5-point Likert scale, and the mean score represented the level of AID. The scale showed good reliability in our data (Cronbach's α = 0.896).

Table 1 Demographic information of the participants

Demographic Characteristics	Category	Frequency	Percentage (%)	AID (*p*)
Gender	Male	267	60.510%	2.355
	Female	175	39.490%	
Grade Level	Freshmen	95	21.440%	1.998
	Sophomores	151	34.180%	
	Juniors	77	17.420%	
	Seniors	119	26.960%	
Age	Under 18 years old	136	30.790%	2.103
	18–22	235	53.080%	
	Over 22 years old	71	16.130%	

Data analysis

This study used IBM SPSS Statistics 24 to conduct normality tests on the variables, guaranteeing the precision and reliability of the findings. Amos 29 software was used

Table 2 The result of data's description and correlation analysis ($n = 442$)

Variables	M ± SD	SK	Kur	TM	CL	FA	TC	AID
TM	3.436 ± 1.0347	-0.862	-1.098	1				
CL	2.882 ± 1.168	0.256	-1.520	-0.364***	1			
FA	2.732 ± 1.240	0.366	-1.596	-0.460***	0.818***	1		
TC	2.709 ± 1.162	0.400	-1.334	-0.391***	0.673***	0.733***	1	
AID	2.851 ± 1.181	-0.036	-1.421	-0.146**	0.631***	0.648***	0.636***	1

***$p < 0.001$

**$p < 0.01$

N, sample size; M ± SD, mean ± standard deviation

MIN minimum value, *MAX* maximum value, *SK* skewness, *Kur* kurtosis, *TM* task motivation, *CL* cognitive load, *FA* future anxiety, *TC* task complexity, *AID* AI dependency

to conduct CFA in order to evaluate the appropriateness and authenticity of the construct validity, ensuring the correspondence between the measurement factors and the items. Taken together, these cross-sectional findings clarify why attention to task characteristics, particularly TC, is theoretically and practically important for understanding AID, while indicating the need for longitudinal or experimental studies to establish directionality. Based on the extensions of SDT and Cognitive Load Theory, the proposed model was tested using Partial Least Squares Structural Equation Modeling (PLS-SEM) using SmartPLS 4 [69]. Using a bootstrap resampling technique with 5,000 resamples, the significance of both direct and indirect effects was evaluated, and 95% bias-corrected confidence intervals were calculated in accordance with the recommendations of Hair et al. [70]. The PLS-SEM method has significant advantages in theory testing and model fitting, making it particularly suitable for research that is both exploratory and predictive [71]. In contrast to conventional structural equation modeling based on covariance (CB-SEM), PLS-SEM is better suited for handling complex model structures and shows greater flexibility when there are tiny sample quantities or data distribution is uneven. Therefore, using PLS-SEM provided a solid framework for this study, ensuring the effectiveness of theoretical validation and empirical analysis.

Results

Descriptive statistics and correlations

The normality of the data was examined using skewness and kurtosis to ensure that the assumptions of multivariate analysis were met. As shown in Table 2, skewness values ranged between – 0.86 and 0.40 and kurtosis values between – 1.60 and – 1.10, all of which fell within commonly accepted cut-offs (|skewness| < 3, |kurtosis| < 10) [72], indicating that the distributions approximated normality.

Table 2 presents the descriptive statistics and intercorrelations for the study variables. On a 1–5 scale, the mean scores were 3.44 (SD = 1.03) for TM, 2.88 (SD = 1.17) for CL, 2.73 (SD = 1.24) for FA, 2.71 (SD = 1.16) for TC, and 2.85 (SD = 1.18) for AID. As expected, TC was positively correlated with CL, FA and AID, and negatively correlated with TM. CL and FA were also strongly and positively related to each other. These patterns are consistent with the hypothesized directions of H1-H4 and provide preliminary support for the proposed mediation model.

Table 3 Gender differences in TM, CL, FA, TC, and AID

	M ± SD		t	*P*
	Male ($n = 267$)	Female ($n = 175$)		
TM	3.467 ± 1.021	3.389 ± 1.056	0.803	0.209
CL	2.916 ± 1.173	2.831 ± 1.161	0.773	0.272
FA	2.733 ± 1.223	2.730 ± 1.269	0.028	0.235
TC	2.733 ± 1.157	2.672 ± 1.172	0.553	0.797
AID	2.954 ± 1.166	2.694 ± 1.189	2.355	0.610

TM task motivation, *CL* cognitive load, *FA* future anxiety, *TC* task complexity, *AID* AI dependency

Group difference analysis

To examine the exploratory hypothesis H5, independent-samples t-tests were conducted to evaluate potential gender differences in AID and its task-related antecedents (Table 3). The analyses revealed no statistically significant gender differences in TM, CL, FA, TC or AID (all $p > 0.05$), although the mean level of AID was slightly higher among male students (M = 2.95, SD = 1.17) than among female students (M = 2.69, SD = 1.19). One-way ANOVAs further showed no significant differences across grade levels or age groups for any of the main variables (Tables 4 and 5), suggesting that the focal associations are relatively stable across demographic subgroups.

Measurement model

Structural reliability and validity

The reliability and convergent validity of the measurement model were first examined (Table 6). All standardized factor loadings exceeded 0.70, indicating that the items loaded strongly on their intended constructs [73]. Cronbach's alpha coefficients ranged from 0.893 (CL) to 0.962 (TM), and composite reliability (CR) values ranged from 0.923 to 0.965, all above recommended thresholds [74, 75]. Average variance extracted (AVE) values were between 0.635 and 0.758, exceeding the 0.50 criterion

Table 4 Grade level differences in TM, CL, FA, TC, and AID

	M ± SD				F	*P*
	Freshmen (*n* = 95)	Sophomores (*n* = 151)	Juniors (*n* = 77)	Seniors (*n* = 119)		
TM	3.541 ± 0.098	3.369 ± 0.084	3.495 ± 0.114	3.400 ± 0.092	0.712	0.545
CL	2.889 ± 0.114	2.815 ± 0.093	3.033 ± 0.128	2.8642 ± 0.105	0.645	0.586
FA	2.755 ± 0.125	2.677 ± 0.096	2.824 ± 0.139	2.724 ± 0.111	0.268	0.848
TC	2.796 ± 0.121	2.647 ± 0.091	2.712 ± 0.124	2.715 ± 0.103	0.340	0.796
AID	2.933 ± 0.116	2.749 ± 0.096	3.095 ± 0.122	2.759 ± 0.105	1.998	0.113

TM task motivation, *CL* cognitive load, *FA* future anxiety, *TC* task complexity, *AID* AI dependency

Table 5 Age differences in TM, CL, FA, TC, and AID

	M ± SD			F	*P*
	Under 18 years old (*n* = 136)	18–22 (*n* = 235)	Over 22 years old (*n* = 71)		
TM	3.521 ± 0.079	3.402 ± 0.068	3.387 ± 0.123	0.712	0.491
CL	2.962 ± 0.097	2.876 ± 0.073	2.750 ± 0.138	0.829	0.437
FA	2.748 ± 0.105	2.760 ± 0.079	2.611 ± 0.137	0.439	0.645
TC	2.760 ± 0.098	2.767 ± 0.074	2.418 ± 0.120	2.851	0.059
AID	2.931 ± 0.103	2.881 ± 0.072	2.603 ± 0.135	2.103	0.123

TM task motivation, *CL* cognitive load, *FA* future anxiety, *TC* task complexity, *AID* AI dependency

[76]. Together, these results suggest satisfactory internal consistency and convergent validity for all constructs.

Discriminant validity

Discriminant validity was evaluated using both the Fornell–Larcker criterion and the heterotrait–monotrait ratio (HTMT) [77]. As shown in Table 7, the square roots of the AVE values for each construct (0.797–0.870) were greater than the corresponding inter-construct correlations, which meets the Fornell–Larcker criterion [76, 78]. In addition, all HTMT values were below the recommended cut-off of 0.90 (Table 8) [76, 79]. Taken together, these results indicate that the five latent variables are empirically distinct.

Confirmatory factor analysis (CFA)

A confirmatory factor analysis (CFA) was conducted to further assess the overall fit of the measurement model. The fit indices indicated acceptable model fit (Table 9): the chi-square to degrees of freedom ratio (CMIN/DF) was 2.935, the RMSEA was 0.064, and the CFI and TLI were 0.922 and 0.915, respectively. All values met commonly used criteria (CFI and TLI > 0.90, RMSEA < 0.08, CMIN/DF < 5) [73, 80], supporting the adequacy of the measurement model.

Multicollinearity test

This study performed a comprehensive diagnostic of multicollinearity between the independent variables to ensure the accuracy of the interpretation and the stability of the model (Table 10). Generally speaking, each variable's variance inflation factor (VIF) should be smaller than 3.3 [81]. All of the variables in this investigation had VIF scores ranging from 1.000 to 3.102 indicate that multicollinearity is not a problem.

Table 6 Reliability and validity of the model

Constructs	Items	Loadings	Cronbach's alpha	CR	AVE
TM	TM1	0.847	0.962	0.965	0.635
	TM2	0.789			
	TM3	0.753			
	TM4	0.833			
	TM5	0.835			
	TM6	0.826			
	TM7	0.773			
	TM8	0.744			
	TM9	0.751			
	TM10	0.785			
	TM11	0.765			
	TM12	0.819			
	TM13	0.784			
	TM14	0.823			
	TM15	0.831			
	TM16	0.780			
CL	CL1	0.910	0.893	0.926	0.758
	CL2	0.875			
	CL3	0.830			
	CL4	0.865			
FA	FA1	0.846	0.920	0.940	0.757
	FA2	0.875			
	FA3	0.857			
	FA4	0.886			
	FA5	0.884			
TC	TC1	0.819	0.962	0.965	0.635
	TC2	0.836			
	TC3	0.871			
	TC4	0.813			
	TC5	0.826			
AID	AID1	0.762	0.896	0.923	0.707
	AID2	0.852			
	AID3	0.879			
	AID4	0.848			
	AID5	0.859			

TM task motivation, *CL* cognitive load, *FA* future anxiety, *TC* task complexity, *AID* AI dependency

Table 7 Fornell-Larcker criterion

Variables	AID	TM	TC	FA	CL
AID	***0.841***				
TM	-0.220	***0.797***			
TC	0.650	-0.440	***0.833***		
FA	0.666	-0.521	0.741	***0.870***	
CL	0.648	-0.412	0.679	0.815	***0.870***

The square root of the AVE values is displayed in bold and italics on the diagonal of the table

TM task motivation, *CL* cognitive load, *FA* future anxiety, *TC* task complexity, *AID* AI dependency

Table 8 Heterotrait-Monotrait criterion

Variables	AID	TM	TC	FA	CL
AID					
TM	0.171				
TC	0.716	0.421			
FA	0.719	0.492	0.813		
CL	0.711	0.384	0.752	0.896	

TM task motivation, *CL* cognitive load, *FA* future anxiety, *TC* task complexity, *AID* AI dependency

Table 9 Model fit indices

Fit index	Final model
CMIN/DF	2.935
RMSEA	0.064
TLI	0.915
CFI	0.922

Table 10 Multicollinearity test

Variables	TM	CL	FA	TC	AID
TM		1.000	1.000	1.000	1.388
CL					3.102
FA					3.028
TC					2.329
AID					

TM task motivation, *CL* cognitive load, *FA* future anxiety, *TC* task complexity, *AID* AI dependency

Table 12 Explanatory power of the model

Constructs	R^2	Q^2
AID	0.545	0.044

AID AI dependency, R^2 represents the explanatory power of the mode, Q^2 represents the predictive relevance of the model

Structural model

Path hypotheses

The structural model was then used to test the hypothesized associations among TC, the mediators and AID (Table 11). TC showed a positive association with AID (β = 0.328, $p < 0.001$) and was strongly related to higher CL (β = 0.684, $p < 0.001$) and FA (β = 0.746, $p < 0.001$), while being negatively associated with TM (β = -0.452, $p < 0.001$). In turn, CL (β = 0.227, $p < 0.001$) and FA (β = 0.337, $p < 0.001$) were positively related to AID, whereas TM was negatively related to AID (β = -0.187, $p < 0.001$). These findings are consistent with the hypothesized directions and provide overall support for the proposed structural model.

Explanatory power and predictive relevance

Using R^2 and Stone-Geisser's Q^2, the model's explanatory power and predictive relevance were evaluated (Table 12). Following Chin and Marcoulides [82], R^2 values of around 0.19, 0.33 and 0.67 can be interpreted as weak, moderate and substantial, respectively. In the present study, the predictors jointly accounted for 54.5% of the variance in AID (R^2 = 0.545), indicating a moderate-to-substantial level of explained variance. In addition, the Stone-Geisser Q^2 value for AID was greater than zero (Q^2 = 0.044), which is commonly taken as evidence of acceptable predictive relevance [83].

Mediation analysis

Using PLS-SEM with bootstrapping, the mediating roles of TM, CL, and FA in the relationship between TC and AID were investigated [84]; the main results are summarized in Table 13 and illustrated in Fig. 2. All three indirect associations between TC and AID were statistically significant, with indirect effect estimates of 0.084 via TM ($p < 0.001$), 0.155 via CL ($p < 0.001$) and 0.252 via FA

Table 11 Path testing results

Hypothesis	β	2.50% CI	97.50% CI	T	*P*	Results
TC → AID	0.328	0.216	0.436	5.858	< 0.001	Supported
TC → TM	-0.452	-0.508	-0.399	15.970	< 0.001	Supported
TC → FA	0.746	0.701	0.785	34.890	< 0.001	Supported
TC → CL	0.684	0.633	0.730	27.434	< 0.001	Supported
TM → AID	-0.187	-0.131	-0.242	6.654	< 0.001	Supported
FA → AID	0.337	0.214	0.456	5.390	< 0.001	Supported
CL → AID	0.227	0.102	0.343	3.710	< 0.001	Supported

TM task motivation, *CL* cognitive load, *FA* future anxiety, *TC* task complexity, *AID* AI dependency

Table 13 Mediation analysis

Relationship	Indirect Effect	Direct Effect	*P*	Results	VAF	Mediation Type
TC → TM → AID	0.084	0.328	< 0.001	Supported	12.90%	PM
TC → CL → AID	0.155	0.328	< 0.001	Supported	23.81%	PM
TC → FA → AID	0.252	0.328	< 0.001	Supported	38.71%	PM

PM partial mediation, *NM* no mediation, *VAF* variance accounted for, representing the proportion of the total effect that is mediated through the indirect path, *TM* task motivation, *CL* cognitive load, *FA* future anxiety, *TC* task complexity, *AID* AI dependency

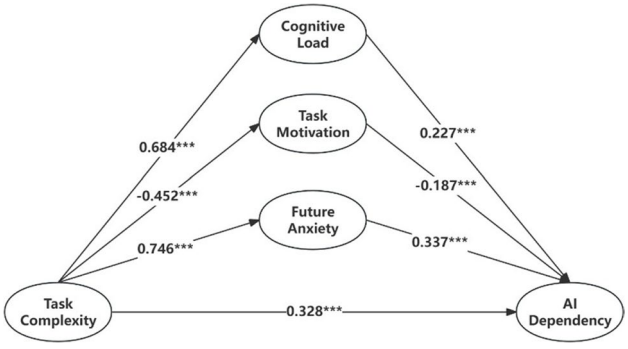


Fig. 2 Mediating effects model results

($p < 0.001$). The direct path from TC to AID also remained statistically significant ($\beta = 0.328$, $p < 0.001$), indicating that the mediators account for part, but not all, of this association. The corresponding variance accounted for (VAF) values were 12.90%, 23.81% and 38.71% for TM, CL and FA, respectively, suggesting that each mediator makes a meaningful contribution to explaining how TC is related to AID, without fully accounting for the association.

Discussion

The purpose of this study was to investigate how TC and AID are related among college students, and to further examine the mediating roles of CL, FA, and TM in this relationship. SmartPLS4 software was used to evaluate the validity of the suggested research model, and all of the study's hypotheses were validated. Below is a thorough analysis of the research findings in relation to the original research questions and hypotheses.

The results indicate that college students' TC is significantly and positively associated with AID; therefore, H1 is supported. Lu and Liu [37] provide support for this interpretation. Their study showed that in applied learning contexts involving AI-related tasks, students were encouraged to engage with AI technologies to manage task demands and develop practical competencies. This pattern suggests that greater TC naturally promotes increased AI use, which indirectly supports our findings on AID. In this study, on the one hand, when college students are required to complete learning tasks involving processing a large amount of information, integrating multiple steps, and evaluating multiple alternatives, they often experience increased pressure and uncertainty about the task path, which makes them more inclined to rely on AI tools to obtain information integration and planning support, thereby improving efficiency and alleviating uncertainty. On the other hand, within the highly digitalized environment of Chinese universities, where academic competition is intense and AI tools are widely integrated into learning platforms and frequently used among peers, digital-native students tend to be more familiar with and trusting of technology, which makes them more likely to depend on AI when dealing with complex tasks. Based on these findings, educators should design tasks so that complexity is moderate and CL is supported, thereby enabling students to handle complex tasks without over-reliance on technological assistance [85, 86]. By providing task decomposition and stage-based guidance, they can help college students gradually improve their ability to handle complex tasks independently and avoid excessive reliance on technological support.

The results show that CL partially mediates the relationship between TC and AID; therefore, H2 is supported. Although there is a lack of direct evidence supporting this pathway, existing studies provide indirect support. Zeitlhofer et al. [43] reported that higher TC is associated with increased germane CL and enhanced meta-awareness, suggesting that learners need to allocate more cognitive resources when engaging in demanding tasks. Similarly, Doger and Gocen [42] found that AI tools can effectively reduce cognitive processing demands during information integration and task planning, indicating that learners may be more likely to rely on AI assistance when experiencing higher CL. Together, these findings imply a possible psychological mechanism in which limited cognitive resources under complex tasks are associated with a greater tendency to use external tools to compensate for processing demands. This interpretation aligns with cognitive load theory and provides a theoretical basis for understanding the mediating role of CL in the present study. In this study, on the one hand, when college students are required to integrate a large amount of information within a limited time, frequently shift their attention, and maintain a high level of focus, their mental resources are more easily depleted, leading to experiences of information overload and attention exhaustion [44]. To reduce this cognitive pressure, they are more inclined to rely on AI tools to simplify complex

information processing and assist in completing tasks. On the other hand, academic tasks in Chinese universities generally emphasize comprehensiveness and interdisciplinary integration, requiring students to make rapid judgments across multiple dimensions of information. The accessibility and convenience of AI tools make students more likely to seek technological assistance under high CL. Cognitive load theory [24] provides an explanation for this result: when task demands exceed an individual's cognitive capacity, utilizing external resources becomes an adaptive strategy. Based on this finding, educators should optimize task design and information organization by reducing irrelevant distractions and providing clear operational guidance, thereby helping students maintain learning effectiveness while minimizing unnecessary CL and reducing passive reliance on AI.

The results show that FA partially mediates the relationship between TC and AID; therefore, H3 is supported. Although there is a lack of direct research on FA, relevant studies provide valuable indirect evidence. Jebur et al. [52] reported that when task challenges are high and exceed learners' capabilities, individuals tend to experience greater uncertainty and concern. Considering the concept of FA, such worries about potential outcomes may partly reflect similar psychological processes. Moreover, Ni and Cheung [48] found that learners' anxiety experiences are closely associated with their tendency to rely on AI tools, with higher levels of anxiety being linked to greater dependence on external support. Therefore, it can be reasonably inferred that the association between TC and AID may be partly explained by learners' concerns about future outcomes. In this study, on the one hand, when college students face tasks with high demands, complicated procedures, and unpredictable outcomes, they are more likely to worry about whether they can successfully achieve their learning goals. Such uncertainty intensifies their future-oriented anxiety. To alleviate this sense of uncertainty, they tend to rely more on AI tools to obtain immediate feedback and structured support, which enhances their perceived control over the learning process. On the other hand, Chinese society places strong emphasis on academic achievement and future development, and the high expectations from families and schools further increase students' sensitivity to potential failure. In this cultural ecosystem, digital-native students are more accustomed to using technological means to reduce uncertainty, which amplifies the relationship between TC and AID. These findings suggest that universities should pay attention to students' FA when designing courses and providing academic support. By clarifying task requirements, offering stage-based feedback, and providing visualized learning progress, educators can help students enhance their sense of control over the learning process, thereby reducing technology dependence caused by anxiety.

The results show that TM partially mediates the relationship between TC and AID; therefore, H4 is supported. The findings of this study indicate that TM plays a significant mediating role in the association between TC and AID. Masrom et al. [57] reported that higher TC is associated with increased TM, while Frenkenberg and Hochman [56] found that learners' motivation levels are closely related to their reliance on AI technologies. Taken together, these findings may suggest that the link between TC and AID could be partly explained by TM. In this study, on the one hand, when college students perceive that high effort yields limited rewards in their learning tasks, their interest and perceived value decline, which reduces intrinsic motivation and autonomous engagement, making them more likely to rely on AI tools to complete learning tasks [87]. On the other hand, in a highly digitalized learning environment where AI tools are widely available and easy to access, college students lacking intrinsic motivation are more likely to view technology as a low-cost coping strategy for completing tasks. SDT [26] supports this finding: when students experience reduced autonomy and diminished competence in their tasks, intrinsic motivation decreases, and they tend to rely more on external tools to compensate for this motivational deficit. This finding suggests that instructors should focus on enhancing the interest and value of tasks when designing courses. By providing positive feedback, competence support, and opportunities for autonomy, educators can strengthen students' intrinsic motivation and help them achieve a healthier balance between technological use and independent learning.

The results show that there are no statistically significant gender differences in AID and its associated psychological variables (CL, FA, and TM); therefore, H5 is not supported. These nonsignificant results may be attributed to two possible factors. First, because current undergraduates are digital natives, both male and female students are highly familiar with AI technologies and their integration into academic platforms. When completing learning tasks involving "processing a large amount of information", "integrating multiple procedural steps", and "evaluating multiple options", both groups tend to rely on AI tools in similar ways to obtain planning support and structured feedback, leading to comparable levels of AID. Second, the cultural and educational context of Chinese higher education may also contribute to the absence of significant gender differences. Within a highly digitalized learning environment characterized by strong performance expectations and intense peer competition, both male and female students are exposed to similar academic pressures and shared social norms [88], which

may result in comparable cognitive, emotional, and motivational responses when interacting with AI-assisted tasks. Given the exploratory nature of this hypothesis and the nonsignificant results, interpretations should remain cautious. Future studies could employ larger and more diverse samples or adopt longitudinal designs to further investigate whether subtle gender-related patterns emerge in AID and its psychological correlates. Nevertheless, these findings suggest that educators should consider developing flexible and personalized AI-assisted learning supports that adapt to students' task-related experiences and engagement preferences, rather than assuming pronounced gender-based distinctions.

Beyond potential gender differences, our findings also raise broader questions about how AI should be positioned in human-centered educational practice. Taken together, these findings suggest that AID among college students emerges at the intersection of task demands, cognitive and emotional resources, and motivational states, rather than being a purely technological phenomenon. This perspective is consistent with broader debates on how AI should be positioned in human-centered practice across different domains. In healthcare, for example, Triberti et al. [89] argue that AI systems can enhance efficiency and decision support, but they also warn that poorly designed implementations may undermine professional autonomy, human relationships, and users' sense of agency. Similarly, in the mental health field, Sebri et al. [90] report that professionals recognize the potential usefulness of AI-based tools, yet remain cautious because current systems lack empathy, emotional understanding, and authentic therapeutic presence. Building on these insights, our results indicate that, in higher education, AI might be most constructively positioned as a scaffold that supports students in managing TC and CL, rather than as a full substitute for autonomous engagement with learning tasks. AI-assisted learning environments could, for instance, provide stepwise guidance, reflection prompts, and strategy suggestions that support self-regulation and TM, while deliberately preserving opportunities for students to generate ideas, make judgments, and monitor their own understanding. Such an approach would leverage the advantages of AI to alleviate excessive burden and anxiety, while at the same time reducing the likelihood that high task demands, strong future-oriented worries, and low motivation are associated with habitual AID in academic work.

Implications

Given the rapid diffusion of AI in higher education, the present findings carry several implications at both theoretical and practical levels. By focusing on TC and its psychological correlates, this study moves beyond individual traits or general technology acceptance and highlights how specific task-related experiences are associated with college students' AID.

At the theoretical level, the results extend Cognitive Load Theory and Self-Determination Theory (SDT) to an AI-supported academic context. The findings indicate that TC is related not only to students' CL and FA but also to their TM, and that these three variables were jointly associated with higher levels of AID. This multipathway pattern enriches existing work that has often treated AI use as a function of stable dispositions or perceived usefulness alone, by underscoring that situational task demands and momentary cognitive–emotional states are also important for understanding students' reliance on AI tools. By identifying CL and FA as correlates of AID, the model also points to psychological processes through which demanding learning environments may be linked to both heightened stress and stronger tendencies to turn to AI as a coping-oriented resource. The integrated framework therefore offers a more nuanced account of AID that links task demands, internal psychological processes, and technology-related behavior within the same model, and provides a basis for future studies to further refine the roles of CL, future-oriented worry, and TM in AI-supported learning.

At the practical level, the findings suggest that educators and institutions should pay attention not only to whether students use AI tools but also to the task conditions under which stronger dependence on AI is more likely to emerge. For instructors, designing assignments with appropriate levels of complexity, scaffolding information-processing demands, and providing clear criteria may be associated with lower CL and FA and, in turn, with a reduced tendency to rely heavily on AI to cope with academic tasks. At the same time, instructional strategies that support more autonomous forms of TM, such as emphasizing task value, offering meaningful choices, and giving constructive feedback, may encourage students to use AI as a complementary resource rather than a substitute for their own thinking. From a mental health perspective, such task designs may help to contain chronic FA and stress related to academic demands, which are often present when students turn to AI as a primary way of handling difficult coursework.

For universities, the results point to the importance of developing guidelines and support services that help students balance the efficiency of AI-assisted learning with the development of independent learning skills, self-regulation, and critical thinking. In addition, campus counseling and mental health services can incorporate brief assessment of AI-related coping and task-related FA into routine psychological screening, and pay particular attention to students who report both elevated FA and a strong tendency to rely on AI to handle academic demands. In such cases, AI use may function as a way of managing

pre-existing academic stress rather than as a neutral learning tool, and may indicate that some students could benefit from strengthened coping resources. Psychoeducational and counseling programs can therefore focus on reducing academic stress and FA, strengthening adaptive coping skills (e.g., planning, time management, problem-solving, and help-seeking), and discussing balanced AI use as one of several available resources. This approach may help support students' emotional well-being alongside their academic development, while avoiding pathologizing typical AI use and instead drawing attention to patterns of reliance that are associated with heightened stress and worry.

Limitations and future research

Although this study offers insightful information about how TC is associated with AID and the mediating roles of CL, FA, and TM, there are several limitations. First, focusing on full-time undergraduate students recruited from multiple universities provided a relatively homogeneous educational stage with comparable course structures and assessment regimes, thereby strengthens interpretability for task-related processes (CL, FA, TM) and reduces between-group heterogeneity. At the same time, postgraduate cohorts and institutional employees (faculty/administrative staff) were not included, so level- and role-based heterogeneity was not examined. Future research may extend this work by adopting a stratified, multi-cohort design that samples across student levels and institutional roles, plans a priori power for multi-group comparisons, and incorporates measurement-invariance tests to ensure comparability across groups. Second, although this study primarily used a quantitative method through questionnaires, which has the advantage of collecting large amounts of data, the quantitative survey may not fully capture the subtle psychological changes and behavioral responses of participants when facing tasks. Therefore, future research could consider combining qualitative methods, such as interviews or focus group discussions, to gain a more comprehensive understanding of students' TC and AID in different context. Third, discipline-specific analyses (e.g., liberal arts, sciences, engineering) were not conducted because participating universities used heterogeneous major taxonomies, which limited consistent cross-site classification. Future studies are encouraged to apply standardized discipline frameworks and ensure adequate subgroup sample sizes and statistical power to examine potential differences in AID and in the pathways from TC through CL, FA, and TM to AID. Lastly, the dynamic process of how TC and AID change over time is not shown by this study because it is based on cross-sectional data. Future studies may adopt longitudinal and experimental approaches to better capture the long-term variations between these variables, thus providing more accurate and targeted strategies for educational interventions.

Conclusion

The current study investigated the possible mediating roles of CL, FA, and TM in the link between TC and AID. The findings indicate that TC is positively associated with AID and may also be indirectly linked to AID via its associations with CL, FA, and TM. By examining these potential mediating mechanisms, the study contributes to the theoretical understanding of how task-related cognitive and motivational factors may relate to AID. Additionally, exploratory analyses revealed no significant gender differences in AID or its task-related psychological correlates (CL, FA, TM), suggesting that demographic variables might play a less central role compared with task-driven cognitive and emotional factors. These findings may offer a conceptual basis for future educational practices aimed at balancing AI usage and autonomous learning and provide new perspectives for psychological research and potential strategies to enhance students' self-regulated learning while mitigating excessive reliance on AI. Taken together, these cross-sectional findings clarify why attention to task characteristics, particularly TC, is theoretically and practically important for understanding AID, while indicating the need for longitudinal or experimental studies to establish directionality.

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Authors' contributions

Conceptualization: Hanhui Li, Jinpeng Liu; Methodology: Hanhui Li; Formal analysis and investigation: Hanhui Li; Writing - original draft preparation: Hanhui Li; Writing - review and editing: Hanhui Li; Supervision: Fan Yang. All the authors have read and agreed to the published version of the manuscript.

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Data availability

The data that support the findings of this study are available on request from the corresponding author.

Declarations

Ethics approval and consent to participate

The researchers confirms that all research was performed in accordance with relevant guidelines/regulations applicable when human participants are involved (e.g., Declaration of Helsinki or similar). This study was approved by the Ethics Committee of Fuzhou University of International Studies and Trade. The participants received oral and written information and provided written informed consent before participating in the study.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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